

HUGO HU

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EDUCATION

Stony Brook University

May 2030

- B.S. Candidate in Computer Science
- Honors College | Honors in Computer Science Program

Stuyvesant High School

June 2026

- Excellence in Mathematics Award GPA: 97.64/100
- Clubs: Stuyvesant Math Team, Mu Alpha Theta Math Honor Society, Spectator / Indicator Photos
- Relevant Coursework: AP Physics C (Mech & E/M), AP Calculus BC, Linear Algebra, Cybersecurity

EXPERIENCE

Memorial Sloan Kettering Cancer Center

New York, NY

Student Intern, Medical Physics

May 2025 - November 2025

- Trained and improved a CNN/LSTM deformable image registration neural network, aligning variable-timepoint cone-beam CT scans to a planning CT reference for lung cancer treatment monitoring
- Experimented with a multi-head attention architecture for the same task, achieving performance comparable to CNN/LSTM neural network and VoxelMorph baselines
- Wrote MATLAB and Python preprocessing and testing code for training datasets
- Ran training and inference on HPC clusters, and configured CUDA environments for compatibility

Summer Intern

July 2024 - August 2024

- Wrote C# and Python data preprocessing code for head/neck cancer CT segmentation training for automatic radiotherapy treatment tracking via MSK AWARE
- Gained familiarity with HPC interfacing and IBM LSF protocols

Hack Club

Shelburne, VT and Remote

Logistics and Hardware Engineering Intern

July 2021 - March 2024

- Built a mail address database implementing USPS Intelligent Mail Barcode (IMb) for no-cost letter tracking, introducing discounted commercial postage pricing access
- Developed low-cost mailing capability, reducing domestic costs by 40% and int'l costs by 80%
- Provided technical expertise and design reviews to dozens of HS students creating their first PCB
- Designed open-source project designs, including a 2D CNC motor control board, USB-C Power Delivery power supply unit, and handheld battery-powered gaming console
- Published open-source reference designs and documentation for NFC tags and USB hubs
- Responsible for the manufacture and mailing of over a thousand hardware kits, shipped globally
- Helped organize in-person hackathons in New York City, Vermont, and San Francisco

PROJECTS

Personal Site Platform | NextJS, Supabase, Cloudinary CDN | [GitHub](#)

2026

- **Personal Site and Photography Portfolio:** Built a personal site with a dynamically-updating photography portfolio, with a NextAuth-secured admin dashboard and shared SQL query APIs
- **Component Inventory Platform:** Built an inventory manager for electronic components using shared UI interface and SQL query APIs, providing a secured login portal, PDF asset label generation, barcode scanner input, and line item component search and modify capability

SAMD21 Corginator | KiCAD, CircuitPython | [GitHub](#)

2024

- Designed and built a corgi-shaped microcontroller development board with the Microchip AT-SAMD21 chipset, compatible with Arduino IDE and CircuitPython, with e-marked USB-C support